

Band broadening

- Evaluate the van Deemter equation which describes the effect of the column and flow rate on the plate height (and therefore on column efficiency)
- Recognize factors that affect the width of a chromatographic peak:
 - linear velocity of mobile phase
 - diffusion coefficient in mobile phase
 - diffusion coefficient in the stationary phase
 - diameter or packing particles
 - thickness of liquid coating
 - retention factor

Why Bands Spread?

Theoretical peak widths can be described in terms of a standard deviation which has several contributing components:

$$\sigma^2_{\text{obs}} = \sigma^2_{\text{injector}} + \sigma^2_{\text{column}} + \sigma^2_{\text{detector}}$$

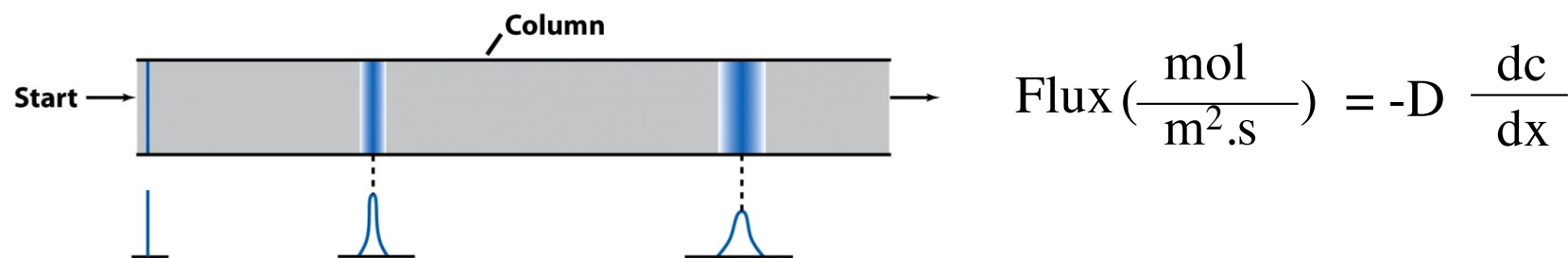
Column broadening factors will be considered as expressed by, H, the height equivalent to a theoretical plate.

$$H = \sigma^2/L = \text{height equivalent to a theoretical plate}$$

... a small H gives the best resolution.

Why Bands Spread?

Fact: Bandwidth is proportional to $\sqrt{t}$: if the elution time is increased by a factor of 4, the band will broaden by a factor of 2



Reference Harris chapter 23

Figure 23-11
Quantitative Chemical Analysis, Seventh Edition
© 2007 W. H. Freeman and Company

$$c = \frac{m}{\sqrt{4\pi Dt}} e^{-x^2/(4Dt)}$$

Concentration profile of m moles diffusing a distance x away from the center.

Compare to equation of Gaussian curve:

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

$$\sigma = \sqrt{(2Dt)}$$

Factors Affecting Resolution: putting it all together

$$R = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right) \left(\frac{k_2}{k_2 + 1} \right)$$

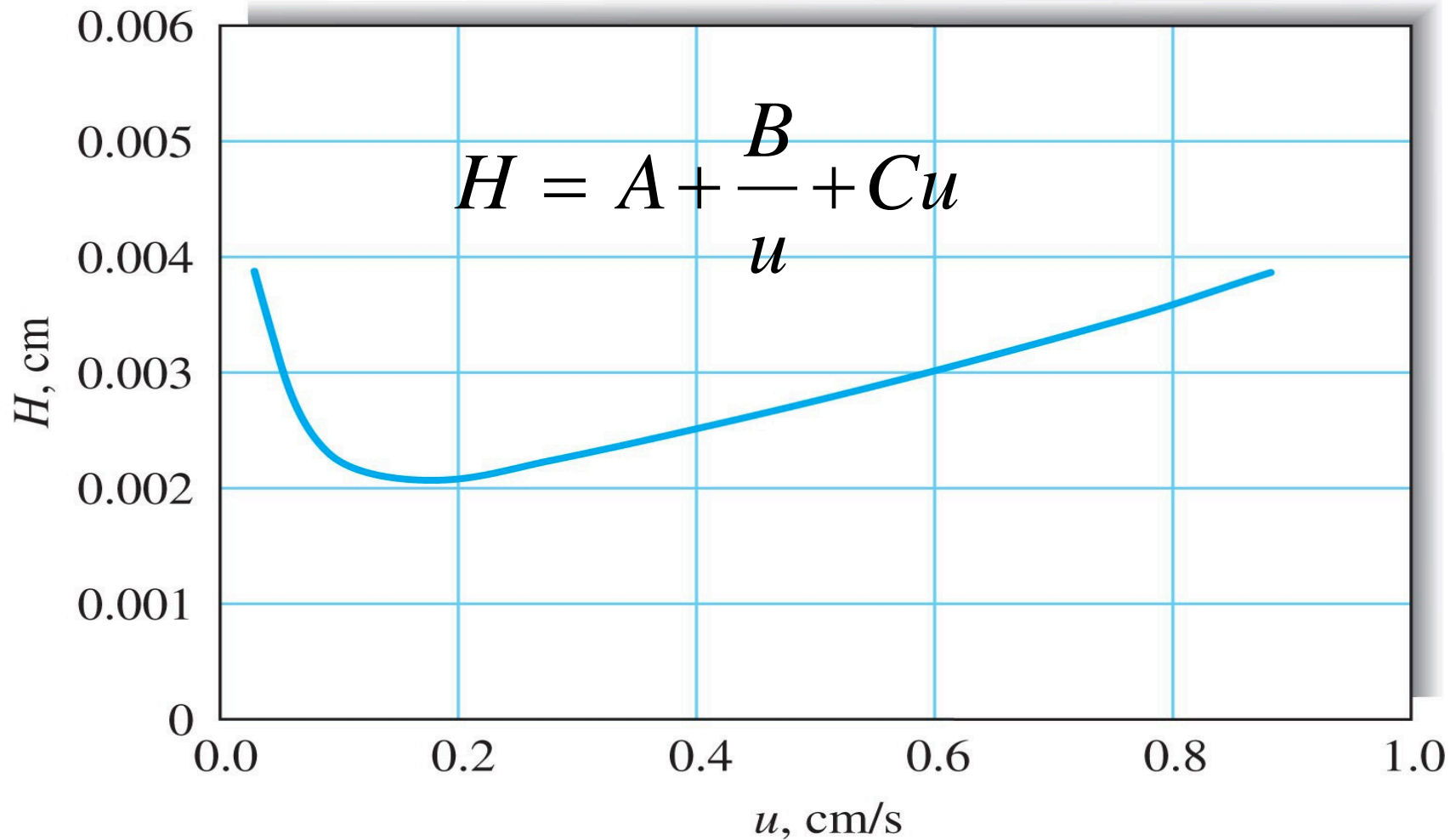
Affect resolution by changing k , α , or N

$$k = \frac{V_s}{V_m} K \quad \alpha = K_2/K_1$$

Increase N either by increasing the length of the column, or by reducing H .

$$N = L/H \quad H = A + \frac{B}{u} + Cu$$

van Deemter Equation for Plate Height



(a) Liquid chromatography

van Deemter Equation for Plate Height

Measure H for several flow rates, F:

1/Flow rate	Flow rate	H	C	B	A
=1/F1	F1	H1	s-C	s-B	s-A
=1/F2	F2	H2	etc.	etc.	etc.
=1/F3	F3	H3			
etc.	etc.	etc.			

- type flow rates in one column
- type H in an adjacent column
- calculate 1/Flow rate to the left of the flow rate
- highlight a 3 col x 5 row matrix
- type =Linest into the upper left cell of the matrix
- select col of H values as y's
- select the 2-col set of x values
- press control-shift-enter to get the coefficients, etc.

Optimum F at $H_{\min}$:

$$H = A + \frac{B}{F} + C F$$

$$\frac{dH}{dF} = 0 - \frac{B}{F^2} + C^{\text{set}} = 0$$

$$F_{\text{opt}} = \sqrt{\frac{B}{C}}$$

(F = volume flow rate)

van Deemter Equation for Plate Height

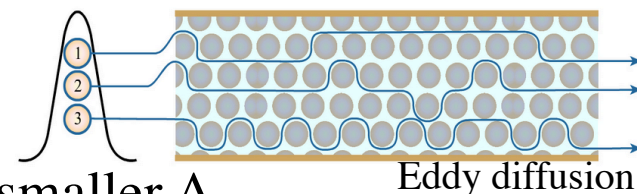
B-term: Longitudinal Diffusion

$$B = 2 D_m$$

D_m = component diffusion rate constant in mobile phase.

The B-term dominates at slow flow rates

$$H = A + \frac{B}{u} + Cu$$



A-term: Multiple Flow Paths

- A depends on particle size; smaller particles give a smaller A.
- A is nearly zero for open tubular (OT) columns.
- At moderate to high flow rates, the A-term is independent of flow rate.

C-term: mass transfer terms

$$C = C_m + C_s$$

C_m : a component to C in the mobile phase

C_s : a component to C in the stationary phase.

The C term dominates a high flow rates.

van Deemter Equation for Plate Height

Mass transfer coefficient, C_s , is directly proportional to the square of the thickness of the stationary phase film and inversely proportional to the diffusion of the solute in the stationary phase, D_s :

$$C_s \propto \frac{d_f^2}{D_s}$$

Mass transfer coefficient, C_M , is directly proportional to the square of the particle diameter in a packed column, d_p , (or column diameter, d_c , in OT column) and inversely proportional to the diffusion of the solute in the mobile phase, D_M :

$$C_M \propto \frac{d_p^2}{D_M} \text{ (packed column)}$$

$$C_M \propto \frac{d_c^2}{D_M} \text{ (OT column)}$$

<https://www.chromacademy.com/channels/hplc-training-courses/principles/hplc-band-broadening/>

Contributions of van Deemter to developments in the field:

- Moving from packed GC to capillary OT
 - Advantage? (consider factors in VanDeemter equation).
 - Why are OT columns, small diameter, capillary, columns?
- Moving from GC to HPLC
 - How does changing from gas to liquid mobile phase impact factors in the van Deemter equation?
 - Why are particle sizes in HPLC packed columns smaller than those in GC columns?

Open tubular column

0.1-0.5mm id, 0.1-5 μ m sp,
thickness, 15-100m length

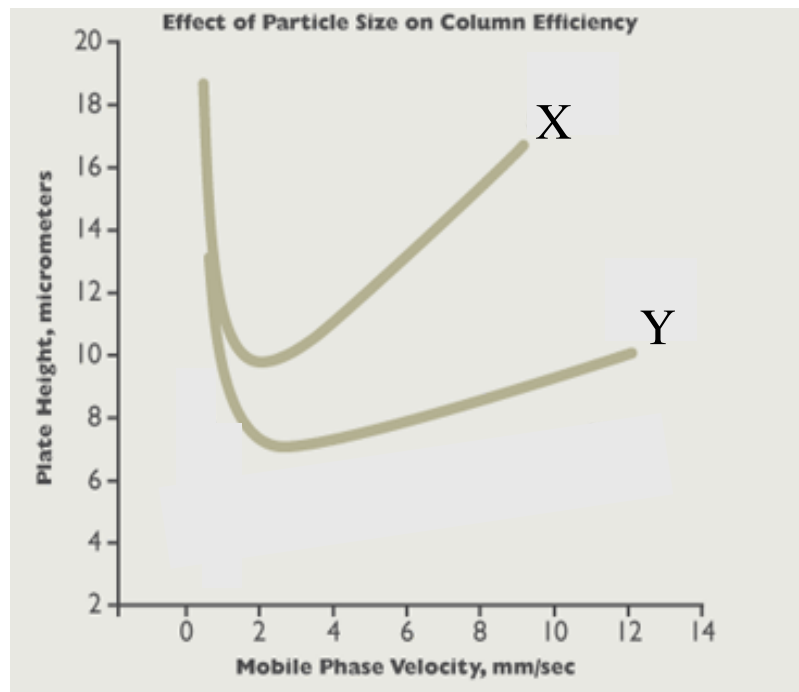
Column made of fused silica
coated with polyimide for
increased thermal and
mechanical stability



Which of the following van Deemter plot results from the smaller particle size HPLC column?

- A. X
- B. Y

$$H = A + \frac{B}{u} + Cu$$



Give one advantage and one disadvantage for using OT rather than packed column in GC.

32-19a: list the variables that lead to band broadening in GC